

Introduction to Analysis Homework 10

May, 15, 2025

1. (7 pt) Let $B = \{f \in C^0([0, 1], \mathbb{R}) : f(x) > 0, \forall x \in [0, 1]\}$. Show that B is an open set in $C^0([0, 1], \mathbb{R})$.
2. (a) (8 pt) Let $f_k : \mathbb{R} \rightarrow \mathbb{R}$ be a sequence of continuous functions that converges uniformly to a function f on $\mathbb{R}$. Prove that

$$\lim_{k \rightarrow \infty} f_k(x_k) = f(x)$$

for every sequence of points x_k with $x_k \rightarrow x$ in $\mathbb{R}$.

- (b) (7 pt) Is the converse of this true?
3. (8 pt) Let $\{f_n\}_{n=1}^{\infty}$ be a sequence of real value continuous functions defined on $[0, 1]$ and assume that $f_n \rightarrow f$ uniformly on $[0, 1]$. Prove or disprove

$$\lim_{n \rightarrow \infty} \int_0^{1-\frac{1}{n}} f_n(x) dx = \int_0^1 f(x) dx.$$

4. Let $f_k : K \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a sequence of functions on a compact set K .
 - (a) (10 pt) Given that f_k is equicontinuous and converging pointwise. Prove that the convergence is uniform.
 - (b) (10 pt) Given that $f_k \in C^0(K, \mathbb{R}^m)$, and f_k converges uniformly on K . Prove that $\{f_k : k \geq 1\}$ is equicontinuous on K .